

# One CSI Report Cannot Serve Every Consumer: The Consumer-Relative False-Clear Rate in 5G/6G Link Adaptation

Andrew H. Bond

Department of Computer Engineering  
San José State University, San José, CA, USA  
andrew.bond@sjsu.edu

**Abstract**—A channel-state-information (CSI) report is a certificate: an implicit claim that the channel is good enough to transmit at a selected rate. That certificate is not read by the channel; it is read by a *consumer*—a scheduler with a reliability target and a decoder at a moment in time. We make the rate at which a CSI certificate *false-clears* (selects a rate the link cannot support) a first-class, measured KPI on a real 5G NR link-level chain (Sionna LDPC decoding), and show it is *consumer-relative* along two axes. First, the reliability target: the identical CSI-derived modulation-and-coding selection that meets an eMBB target ( $\text{BLER} = 10^{-1}$ ) misses a URLLC target ( $10^{-3}$ ) by more than two orders of magnitude ( $112\times$ ), while a target-aware policy meets it. Second, the refresh horizon: as the report ages, the false-clear rate grows, and the largest safe report period is proportional to the Clarke coherence time ( $\approx 0.177 T_{\text{coh}}$ ,  $R^2 = 0.915$ ). The witnessed corrections that hold each axis—outer-loop link adaptation, a coherence-driven refresh—already exist; our contribution is the measured number that says when each is needed. All results are drawn from sealed, pre-registered records.

**Index Terms**—CSI feedback, link adaptation, URLLC, outer-loop link adaptation, reliability, 5G NR.

## I. INTRODUCTION

Adaptive modulation and coding (AMC) turns a channel-state-information (CSI) report into a rate decision: the transmitter selects the highest modulation-and-coding scheme (MCS) whose required SINR the reported CSI clears. The report is thus a *certificate*—an implicit assertion that the link will decode at the chosen rate. Like any certificate, it can be wrong in a way that matters only to whoever reads it. We call a wrong-in-the-optimistic-direction certificate a *false-clear*: the CSI cleared an MCS that the true channel cannot support, and the transport fails (a HARQ NACK, a block error).

Our thesis is that the false-clear rate is not a property of the channel or even of the report alone; it is a property of the report *and its consumer*. Two schedulers reading the same CSI can disagree on whether it false-clears, because they read it against different requirements. We make the false-clear rate a measured KPI on a real 5G NR link-level chain and demonstrate this consumer-relativity along two operationally important axes:

- **the reliability target (§III)**: a single CSI-derived MCS that is adequate for enhanced mobile broadband (eMBB)

is catastrophically optimistic for ultra-reliable low-latency communication (URLLC), by more than  $100\times$ ;

- **the refresh horizon (§IV)**: the false-clear rate grows as the certificate ages, and the safe refresh period scales with the channel coherence time by a simple measured law.

Neither observation requires a new estimator or scheduler. The corrections are known: outer-loop link adaptation (OLLA) [1] closes the target-relativity gap, and coherence-driven CSI reporting closes the refresh gap. What has been missing is the *measured, consumer-relative false-clear rate* that quantifies when each correction is load-bearing and how large the exposure is when it is absent.

## II. METHOD

**Substrate.** We use a real 5G NR link-level chain in Sionna [2]: LDPC-coded QAM over the NR PDSCH, with block error rate (BLER) measured from the actual LDPC decoder rather than an SINR-to-BLER formula. The decoder is the *witness*: it independently determines whether the consumer decoded.

**Certificate, consumer, witness.** An AMC certificate selects the highest MCS whose required SINR the CSI clears at a given BLER target. The *consumer* is that rate decision under a specific reliability target; the *witness* is the LDPC decode outcome. A false-clear is a certificate that clears an MCS the witness then fails.

**Discipline.** Each experiment is pre-registered with numerical bars and manipulation checks fixed before the graded run; the sealed record is content-hashed and graded on seeds disjoint from those used to design it, and negative or failing outcomes are kept [5]. The numbers below are the graded, sealed values.

## III. CONSUMER-RELATIVITY IN THE RELIABILITY TARGET

A base station serving mixed traffic derives one CSI estimate per user and selects an MCS. Consider the same estimate consumed under two targets: eMBB ( $\text{BLER} \leq 10^{-1}$ ) and URLLC ( $\leq 10^{-3}$ ). We drive a real NR link at high mobility, apply the naive CSI-derived MCS, and measure the achieved BLER against each target; we then compare a *target-aware* policy that sizes its margin to the consumer's target.

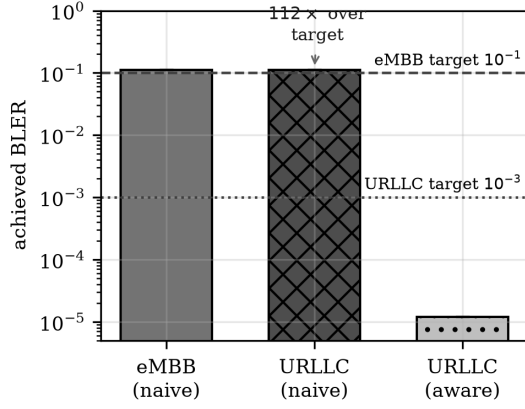


Fig. 1. Consumer-relativity of the CSI certificate in the reliability target (real NR, Sionna LDPC; three sealed seeds). The identical CSI-derived MCS meets the eMBB target ( $10^{-1}$ ) but false-clears the URLLC target ( $10^{-3}$ ) by  $112\times$ ; a target-aware policy meets URLLC at the cost of  $\sim 2$  dB margin.

Figure 1 reports the sealed result (three graded seeds). The naive certificate achieves  $\text{BLER} \approx 0.112$ . Read by the eMBB consumer this is at target—the certificate is sound. Read by the URLLC consumer the *same* 0.112 is a false-clear by  $112\times$  (against  $10^{-3}$ ): every such block is a reliability-SLA violation, invisible to a scheduler that assumed one CSI report serves all traffic. The target-aware policy—the same CSI, an added  $\approx 2$  dB margin sized to the URLLC target—achieves  $\text{BLER} \approx 1.2 \times 10^{-5}$ , comfortably within target, at the cost of that margin’s spectral efficiency. No single margin is correct for both consumers: the eMBB-adequate margin fails URLLC, and the URLLC-safe margin over-provisions eMBB. The false-clear rate is the number that prices the difference.

#### IV. THE REFRESH HORIZON

The second axis is time. A CSI report is fresh when issued and ages as the channel decorrelates; a certificate that was sound at report time false-clears once the channel has moved. At  $f_D = 200$  Hz with a 20-TTI report period, the naive (held-CSI) certificate reaches  $\text{BLER} \approx 0.36$  against a 0.1 target—a false-clear on roughly a quarter of transmissions—while OLLA, the deployed witnessed correction that adjusts an offset from HARQ feedback, holds it at  $\approx 0.10$  (Fig. 2(a) shows the aging across mobility).

The operational question is how often to refresh. Sweeping Doppler and report period, we define the *refresh floor* as the largest report period keeping BLER at target, and find it proportional to the Clarke coherence time  $T_{\text{coh}} = 0.423/f_D$ :

$$\text{refresh floor} \approx 0.177 T_{\text{coh}}, \quad R^2 = 0.915, \quad (1)$$

across an order of magnitude of mobility (Fig. 2(b)). At high Doppler the floor saturates at the minimum report period (one TTI): the certificate cannot be refreshed faster than the frame allows, so above a mobility threshold no reporting cadence suffices and a witnessed (HARQ-driven) correction becomes mandatory rather than optional. Equation (1) gives a concrete

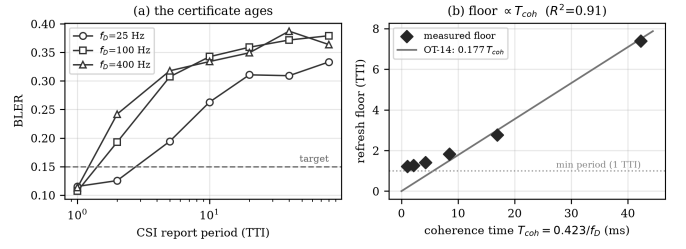


Fig. 2. The CSI certificate’s refresh horizon (real NR, Sionna LDPC; sealed sweep). (a) The certificate ages: BLER grows with report period, faster at higher Doppler. (b) The refresh floor is proportional to the Clarke coherence time ( $0.177 T_{\text{coh}}$ ,  $R^2 = 0.915$ ); at high Doppler it saturates at one TTI.

provisioning rule: re-report before the channel decorrelates past  $\approx 0.18 T_{\text{coh}}$ , not on a fixed calendar.

#### V. RELATED WORK

Link adaptation, CSI aging, and outer-loop correction are mature. OLLA [1] adjusts an SINR offset from HARQ ACK/NACK to hold a BLER target under imperfect or stale CSI; it is precisely the witnessed correction our refresh-horizon result quantifies. Short-packet and URLLC analysis [3] establishes the reliability-target regime in which the eMBB-to-URLLC gap of §III lives. Learned CSI feedback trained on reconstruction error [4] is an active direction; we note only that the reconstruction-versus-consumer objective gap it might imply did not appear at practical operating points in our controlled tests, and we make no claim on it here. Our contribution is orthogonal to all three: not a better estimator, scheduler, or offset rule, but a *measurement*—the false-clear rate as a first-class, consumer-relative KPI—and the two relativities and the refresh law it exposes.

#### VI. DISCUSSION AND CONCLUSION

The unit of correctness in link adaptation is not the CSI report but the pair (report, consumer). Sizing margin and refresh cadence for the consumer’s reliability target and coherence time, rather than for a nominal channel, is what keeps a certificate honest. The false-clear rate makes the cost of ignoring the consumer measurable:  $112\times$  for a mismatched target, a coherence-scaled horizon for a stale report. The Sionna link-level chain is the evidence rung; an over-the-air or srsRAN testbed with real HARQ traces is the external-validity graduation. The methodology is not radio-specific—the same consumer-relative false-clear appears in optical quality-of-transmission and database replication [5]—but the results here stand on the 5G NR CSI certificate alone.

#### REFERENCES

- [1] A. Sampath, P. S. Kumar, and J. M. Holtzman, “On setting reverse link target SIR in a CDMA system,” in *Proc. IEEE VTC*, 1997, pp. 929–933.
- [2] J. Hoydis *et al.*, “Sionna: An open-source library for next-generation physical layer research,” arXiv:2203.11854, 2022.
- [3] G. Durisi, T. Koch, and P. Popovski, “Toward massive, ultrareliable, and low-latency wireless communication with short packets,” *Proc. IEEE*, vol. 104, no. 9, pp. 1711–1726, 2016.

- [4] C.-K. Wen, W.-T. Shih, and S. Jin, "Deep learning for massive MIMO CSI feedback," *IEEE Wireless Commun. Lett.*, vol. 7, no. 5, pp. 748–751, 2018.
- [5] A. H. Bond, "Observation Theory: consumer-relative witnessed certificates," sealed pre-registrations and code, <https://github.com/ahb-sjsu/observation-theory-campaigns>, 2026.